ACM-68000

Agentic Commerce Manifest Protocol

FOUNDER'S DOCUMENT

Operating Design Reference & White Paper

Datalake Operator Grade

First Enterprise-Grade MCP with Security Stack

Allooloo Technologies Corp.

Vancouver, Canada | Barcelona, Spain

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1. THESIS

> *Large Language Models on hyperscaler fabric have known drift. The web they navigate is 90% synthetic slop and 51% bot traffic. Traditional agent-to-agent communication requires interpretation — a liability when instruments are unreliable and the ocean is hostile.*

ACM-68000 solves this by applying the same principles mariners used for millennia: fixed trusted anchors for triangulation against drift, and counterparty verification to confirm signals arrived intact.

The Navigation Metaphor

In celestial navigation, you don't trust the ocean — it moves, it lies, it kills. You trust fixed points in the sky that have been there for millennia. Sailors could cross oceans because Polaris doesn't move.

In shortwave radio and cryptography, you need counterparties — known trusted stations with verified call signs. A signal isn't valid until the receiver confirms it.

ACM-68000 combines both: fixed infrastructure anchors (two geographically separate Azure tenants) that LLMs can triangulate against, plus a receiver station that confirms every signal. The result is deterministic commerce execution in <40ms, with zero interpretation required.

Two Byproducts

1. **Trust anchoring for drifting LLMs:** External fixed points that agents can triangulate against, preventing hallucination and confabulation in commerce decisions.
2. **Elevation above the slop:** Business-critical signals travel via Ghost Headers, invisible to the garbage web, elevated above the noise floor where 90% is synthetic and 51% is automated.

2. PROBLEM STATEMENT

The Three Failures

FAILURE	DESCRIPTION
Hyperscaler Drift	LLMs on cloud fabric hallucinate, lose context, confabulate. The instruments are unreliable. You cannot trust an agent's 'reasoning' for commerce decisions.
Hostile Web	90% of web content is now synthetic. 51% of traffic is automated bots. Agents navigating this landfill are interpreting garbage. Interpretation is a liability.
No Counterparty	Traditional A2A has no verification. Agents 'discuss' purchases, 'reason' about intent. There's no receiver station confirming the signal arrived. No proof of transmission.

Traditional A2A vs. ACM-68000

TRADITIONAL A2A	ACM-68000
Agent 'discusses' purchase	Agent issues machine-executable signal
LLM 'reasons' about intent	Keyword-to-schema mapping (<40ms)
Variable text response	Fixed 7-signal vocabulary
Requires interpretation	Pipes directly to ERP/Dataverse
No audit trail	Every signal logged in InboundEvents
Hope-based execution	CEO dashboard visibility

3. THE SOLUTION: TWO TENANTS, ONE LOOP

Infrastructure Overview

TENANT	ACCOUNT	ROLE	KEY URLS
#1 — Broadcast	…OUTBOUND.onmicrosoft.com	Signals OUT	dpuone.ai, mcp.10060.ai, acm-000.ai - (acm-200.ai, acm-300.ai, acm-403.ai, acm-404.ai, acm-451.ai, acm-5000.ai = 7 combined signals)
#2 — Receiver	…INBOUND.onmicrosoft.com	Signals BACK	7signalingress.ai, aio-router.ai, 7signalrefinery.ai

Tenant #1 — Signal Source / Broadcast

Resource Group: rg-waypoint-10060-fr

Region: Azure France Central (EU sovereign)

Hosts:

- **dpuone.ai** — Digital Proof Unit (GTIN origin, jurisdiction verification)
- **mcp.10060.ai** — MCP Server (first CPG MCP in official registry)
- **acm-000.ai through acm-500.ai** — The 7 Signal nodes

Tenant #2 — Receiver + Security Stack

Resource Group: rg-7signalfoundry-fr

Account: Microsoft AI Partner Program

Hosts:

- **7signalfoundry.ai** — Forges signal infrastructure
- **7signalingress.ai** — Receives agent POST-backs (/ingest endpoint)
- **aio-router.ai** — CEO Dashboard (real-time signal visibility)
- **7signalrefinery.ai** — Enterprise signal processing
- **7signalrefinery.org** — MIT Protocol specification
- **7signalstack.ai** — Protocol standard documentation

Why Two Tenants?

Broadcast ≠ Receiver. Separate trust domains. Like shortwave radio: the transmitting station and receiving station are independent. If Tenant #1 is compromised, Tenant #2 can still validate. If an agent spoofs a signal, the receiver on Tenant #2 won't match. No single point of failure.

4. THE 7 SIGNALS

Every commerce decision resolves to exactly one of seven deterministic states. No interpretation. No reasoning. Just execution.

SIGNAL	STATE	MEANING	COLOR
acm-000.ai	NOT_APPLICABLE	Idle / No state	#888888
acm-200.ai	ALLOW	Execute transaction	#1D9E75
acm-300.ai	CONDITIONAL	Verify / Audit required	#EF9F27
acm-403.ai	RESTRICT	Blocked	#E24B4A
acm-404.ai	NOT_FOUND	Product missing	#444444
acm-451.ai	ESCALATE	Human in the loop	#E24B4A
acm-500.ai	SYSTEM_ERROR	Retry later	#7F77DD

Why 7 Signals?

- **Minimal:** Covers all possible agentic commerce outcomes
- **Universal:** Maps to HTTP status codes developers already understand
- **Deterministic:** No ambiguity, no interpretation required
- **Cacheable:** Can be cached at edge indefinitely
- **Machine-readable:** HTTP headers, not prose

5. AGENTIC GHOST HEADERS

HTTP response headers that every ACM-68000 node returns. Called "Ghost" because:

- **Invisible to humans** (they don't see HTTP headers in a browser)
- **Visible to AI agents** (they read headers programmatically)

The Complete Header Set (15+ Headers)

HEADER	VALUE / PURPOSE
X-GSC-Protocol	ACM-68000
X-GSC-Signal	ACM-200 (current state)
X-GSC-State	ALLOW (human-readable state)
X-GSC-Router	aio-router.ai
X-GSC-Stack	7SignalStack
X-GSC-Operator	Allooloo Technologies Corp.
X-GSC-Node	<node-identifier>
X-GSC-Version	1.2.1
X-GSC-Registry	io.github.allooloo/acm-68000-mcp
X-GSC-Inbound	https://7signalingress.ai/ingest (RETURN ADDRESS)
X-GSC-Trust-Anchor	dpuone.ai
X-GSC-Verify	/.well-known/acm-68000.json
X-GSC-Timestamp	<ISO-8601> (freshness)
X-GSC-Nonce	<unique-request-id> (anti-replay)
X-MCP-Server	https://mcp.10060.ai
X-MCP-Registry	io.github.allooloo/acm-68000-mcp
X-Agent-Access	Claude,OpenAI,Gemini,Mistral,Grok

6. SIGNAL ROUND TRIP

Outbound — Signals OUT (Tenant #1)

```
Agent → mcp.10060.ai → dpuone.ai → Returns 1 of 7 signals + 15 Ghost
Headers + X-GSC-Inbound return address
```

The broadcast tenant sends deterministic signals. Every response includes the return beacon (X-GSC-Inbound) pointing to Tenant #2.

Inbound — Signals BACK (Tenant #2)

```
Agent POSTs to https://7signalingress.ai/ingest → Azure Front Door → WAF →
Dataverse → aio-router.ai
```

The Receiver Station — Technical Flow

This is the breakthrough — the counterparty that completes the cryptographic handshake:

STEP	ACTION
1. INPUT	AI Agent sends: "Need 4 pallets diapers Madrid"
2. PARSE	Parser extracts: cluster=HYGIENE, qty=4, region=ES-MAD, gtin=null
3. RESOLVE	Signal resolver → ACM-300 (CONDITIONAL) because GTIN missing
4. WRITE	OData client POSTs to Dataverse (cr1e4_inboundeventses table)
5. DISPLAY	CEO sees signal in Power Apps dashboard with color coding

The signal lands in the CEO dashboard before a traditional A2A system would finish its first reasoning step. With a target <40ms total latency. FaF principles in play.

7. SECURITY STACK — FIRST EVER FOR MCP

All security infrastructure is deployed on Tenant #2 (the receiver). This is the first MCP with enterprise-grade security.

LAYER	FUNCTION
Azure Front Door Premium	Global load balancing, CDN, edge delivery, entry point for all inbound signals
WAF (Web Application Firewall)	DDoS protection, bot filtering, OWASP ruleset, rate limiting
Managed Identity	No secrets in code — Azure handles authentication
Dataverse	InboundEvents table, row-level security, enterprise audit trail
Power Apps	CEO Dashboard at aio-router.ai, RBAC (Role-Based Access Control)

Trust Model

ANCHOR	PROVIDER	PURPOSE
Azure Paris	Microsoft	EU sovereign hosting, GDPR compliance
Cloudflare	Cloudflare	TLS 1.3, WAF, DDoS protection, edge delivery
Domain Control	Allooloo	Exclusive control of all .ai signal domains

Security Mitigations

RISK	MITIGATION
Domain Spoofing	TLS certificate validation + allowed domain list
Man-in-the-Middle	TLS 1.3 + Cloudflare encrypted tunnel
Replay Attacks	X-GSC-Timestamp + X-GSC-Nonce
Header Injection	Cloudflare WAF strips malicious headers

8. LIVE PROOF — TEST COMMANDS

The following commands can be run right now to verify the system is operational:

Health Check

```
curl https://aio-orderable-h4gfg8bwbyb2d5az.francecentral-
01.azurewebsites.net/health
```

Expected: {"status":"ok","version":"1.3.0","dataverse_configured":true}

Test Ingest Endpoint

```
curl -X POST "https://aio-orderable-h4gfg8bwbyb2d5az.francecentral-
01.azurewebsites.net/ingest" -H "Content-Type: application/json" -d '{"message": "Need 4
pallets diapers Madrid", "sender": "test-agent"}'
```

Expected: {"dataverse_saved": true, "signal": "acm-300.ai", ...}

Check Ghost Headers

```
curl -I https://7signalrefinery.ai
```

Expected: X-GSC-Protocol, X-GSC-Signal, X-GSC-Operator, X-MCP-Server headers present

9. COMPLETE URL REFERENCE

Tenant #1 — Broadcast Infrastructure

URL	ROLE
dpuone.ai	Digital Proof Unit — GTIN origin, jurisdiction verification
mcp.10060.ai	MCP Server — first CPG MCP in official registry
acm-000.ai	NOT_APPLICABLE signal node
acm-200.ai	ALLOW signal node
acm-300.ai	CONDITIONAL signal node
acm-403.ai	RESTRICT signal node
acm-404.ai	NOT_FOUND signal node
acm-451.ai	ESCALATE signal node
acm-500.ai	SYSTEM_ERROR signal node

Tenant #2 — Receiver Infrastructure

URL	ROLE
7signalfoundry.ai	Forges signal infrastructure
7signalingress.ai	Receives agent POST-backs (/ingest endpoint)
aio-router.ai	CEO Dashboard — real-time signal visibility
7signalrefinery.ai	Enterprise signal processing
7signalrefinery.org	MIT Protocol specification
7signalstack.ai	Protocol standard documentation

Signal Flow Order

```
7signalfoundry.ai → 7signalingress.ai → aio-router.ai → 7signalrefinery.ai →
                    7signalrefinery.org → 7signalstack.ai
```

10. CONCLUSION

> *ACM-68000 is the first Model Context Protocol with enterprise-grade security, counterparty verification, and deterministic signal execution. It applies the principles of celestial navigation and shortwave radio to the problem of LLM drift on hyperscaler fabric.*

What We Built

- **Two-tenant trust architecture** — Broadcast (Tenant #1) and Receiver (Tenant #2) are separate trust domains
- **Fixed trusted anchors** — Like shortwave radio stations with known call signs
- **Counterparty verification** — The receiver station (7signalingress.ai) confirms every signal
- **Agentic Ghost Headers** — Invisible to humans, visible to AI agents, machine-readable identity
- **Enterprise security stack** — First MCP with Front Door Premium, WAF, Managed Identity, Dataverse, Power Apps
- **CEO visibility** — Real-time dashboard showing every signal as it lands

The Result

Deterministic commerce execution in targeted <40ms. FaF! Zero interpretation. Zero reasoning. Signals out from Tenant #1, signals back to Tenant #2. Full loop. First ever for MCP.

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github.com/allooloo/acm-68000

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